



Emergence Theory

An Overview





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Overview

Emergence is one of the central concepts within systems and complexity theory as it describes a universal process of becoming or creation, a process whereby novel features and properties emerge when we put elementary parts together as they interact and self-organize to create new patterns of organization. Emergence being a highly abstract concept is literally everywhere, from the evolution of the universe to the formation of traffic jams, from the development of social movements to the flocking of birds, from the cooperation of trillions of cells giving rise to the human body to the formation of hurricanes and financial crises.

Although the ideas of emergence have been of interest to many for millennia it has often been seen as something of a mystery, but with the development of complexity theory, we increasingly have the computational and conceptual tools to understand it in a structured, scientific fashion. During the book, we will be drawing upon different ideas in complexity and systems theory to build up a framework for understanding emergence in a coherent fashion. More specifically, we will explore emergence as a form of nonlinear pattern formation. Where synergies between elementary parts give rise to self-organization and the formation of a distinct pattern, that creates new, emergent levels of organization, that are driven by an evolutionary dynamic.

After giving an overview of emergence theory, the book is designed around four main sections. In the first section, we start off by talking about patterns of correlation in general before going on to look at synergistic interactions that are the foundations to emergence.

The next section is focused on pattern formation, the question of how the parts come to self-organize; to synchronize their states into forming a new level of organization. Here we will talk about the two primary different types of emergence that are often used categorizations; what are called strong and weak emergence.

In the third section, we will look at the idea of integrative levels, how synergies give rise to pattern formation and the emergence of new levels of organization called integrative levels. We will talk about how these different levels come to have their own irreducible internal structure and processes that result in a complex dynamic between the micro and macro levels of organization.

In the last section of the book, we will look at how emergence plays out over time within some process. We will talk about the edge of chaos hypothesis; how self-organizing, emergent systems never quite lock into place but instead evolve through a dynamic interplay between order and disorder, to create novel phenomena at new levels of complexity.

This is an introductory book and is non-technical, however, it is important to note that the concept of emergence is highly abstract, to do it justice we will have to use high-level abstractions, as such students will need to feel comfortable with formal abstract models.

The book should be accessible without need for any specific background in science and should be of relevance to many different domains, in particular for those in the areas of computer science, biology and ecology, philosophy, the cognitive sciences and anyone with an interest in better understanding this central concept with the complexity and systems theory framework.

Emergence

Emergence is a term used in philosophy, art, and science to describe how new properties and features are created as we put things together. Emergence describes a process whereby component parts interact to form synergies, these synergies then add value to the combined organization which gives rise to the emergence of a new macro-level of organization that is a product of the synergies between the parts and not simply the properties of the parts themselves.

Because emergent properties are a produce of the synergies between the parts, they cannot be observed locally in the subsystems, but only as a global structure or integrated network. In such a way emergence creates a system with two or more distinct and irreducible patterns of organization, called integrative levels.

Thus new descriptive categories are necessary for the different levels when referring to phenomena that involve emergence. As emergent macro-level phenomena cannot be described within the vocabulary applicable to the parts; these emergent features require new terms and new concepts to categorize them. As an emergent phenomenon is thought to be irreducible, in some sense, to an account of its elementary parts, a true emergent phenomenon is one for which the optimal means of prediction is computer simulation.

Emergence involves the creation of something new that could not be expected from a description of the parts prior to its creation. This novelty comes in a spectrum of strengths. These strengths account for the different types of emergence, which are described as being strong or weak. At the weak end of the spectrum, the novel phenomena may only appear different, while after their arrival it is possible to understand how they derive from some set of elementary parts. At the strong end of the spectrum, one thing is caused by another but is in no way reducible to the other.

Classical examples of emergence can be found in the collective behavior of social insects like ants and bees as they create swarms and colonies. Many other examples have been identified in human social organizations and human consciousness. The solvent property of water is an often cited example of emergence, for neither hydrogen atoms nor oxygen atoms in isolation possess this property and neither do they possess scaled-down versions of the property. Solvent action seems to emerge from a nonlinear combination of the properties of hydrogen and oxygen.

A primary distinction within the philosophical study of emergency is often made between what is called ontological and epistemological emergence. Epistemology refers to knowledge and how one knows the world. Epistemological emergence is the idea that some systems cannot be described, as a matter of practice, in terms of their component units because of the limits to our knowledge, that is our inability to obtain all relevant information and to do the computations.

Ontology refers to the nature of being, it is concerned with what we consider the objective being of things in the world around us. With ontological emergence, we are not just making statements about our own knowledge of the world we are making statements about how the world actually is. When we talk about ontological emergence we are making a statement about how the world is irrespective of our understanding of it. If ontological emergence is identified it means a full understanding of a system in terms of its components is not possible in theory, not just because of practical constraints, but because new and fundamentally irreducible levels of organization appear at higher levels.

Emergent properties are attributed to whole structured collections of elements, where the emergent property is not an additive function of the properties of the elements of the collection taken individually. For example, the mass of the human body is a simple summation of all of its parts taken in isolation and thus is not an emergent feature. However, human consciousness would appear to be an emergent phenomenon as we are not able to provide an account of it as some summation of elementary cognitive parts. The behavior of the system results more from the interaction of the components than from the behavior of the components themselves. The added value of the whole - that exists above and beyond that of the parts and their properties - is a product of how they interact. These specific interactions between parts that add or subtract value from the whole are called synergies. Emergence is then a product of synergistic interactions between the parts in the system.

Positive synergies arise when two or more elements both differentiate their states and activities with respect to each other and coordinate them. Differentiation enables the parts to specialize, while integration enables them to coordinate their different capabilities towards an overall outcome, in so doing the synergistic interaction adds value to the overall system and we get some combined organization that is greater than the sum of its parts; emergence. These synergies are a pervasive phenomenon in our world, seen in the interaction of pollinating insect and plant; microbial organisms and their hosts; in the interaction between different drugs; and in all kind of social organization such as businesses, families, and friendships.

Because the parts within a synergy are differentiated and coordinated with respect to each other, this means they have a specific type of interaction and if we alter that interaction by moving the parts around or recombining them in a different fashion then the added value may be lost. For example, if we had a lock and a set of keys with only one of these many keys being able to open the lock, although in their mass and other physical properties the keys would be almost exactly the same, if we took any key except the correct one and combined it with the lock this combination would have virtually zero value as the lock would not open and thus the combination would be non-functional. But when we combine the right key with the lock we will get an overall functioning system that is of value. This added value was derived from the synergy between the key and lock, this specific key and lock were designed to be differentiated with respect to each other and coordinated, thus enabling their combined functionality. Neither lock nor key could achieve the combined function of securing something in isolation. Thus we need both different parts but also they need to be working together in some fashion to get the overall functioning system. The other keys were differentiated from this lock but they were not coordinated with it and thus no added value or functionality emerged when we combined them.

Synergistic interactions give rise to new levels of organization that have their own internal properties, features, and dynamics. These new emergent levels are called integrative levels. Although these new levels are not directly dependent upon the properties of their parts, the pattern of organization that has emerged is dependent upon the integrity of the synergies between their constituent parts. If we take the synergies away the integrative levels will disintegrate and disappear. Thus unlike simpler linear systems that can be reduced to a single level of organization, emergence results in the development of more than one level of organization, with these different levels representing a different context and sets of rules governing the parts.

With integrative levels, we get a dynamic where the higher level is dependent upon the lower levels for its existence, however, the higher level also creates its own pattern of organization that feeds back to condition the micro-level elementary parts and interactions. All emergent systems are composed of a micro-level set of building blocks that place an upward set of physical constraints on the system, but also the macro-level defines a pattern of organization that then exerts a downward effect on the parts by creating the context for their operation. For example, the electrons moving through a microprocessor are the physical building blocks that make possible computation. The processor is one of the physical constraints placed on the macro-level software program that is running.

However, the program on the macro-level is performing some process that is not associated with the organization of the parts - such as a person using it to create a website - all though the macro-level operations of the software is fully dependent upon the micro-level of the physical electrons moving around in the processor, the structure of that macro-level process that the person is performing when they build the website has nothing to do with the underlying micro-level physical processing going on in the computer. This software activity could be conducted in the same way on many different types of computers with very different underlying patterns in the hardware, the upper level of the software is through creating the downward effect conditioning how the electrons move around on the micro-level.

Due to the development of different levels of organization within a single overall system, emergence gives rise to a complex dynamic between the different levels; most notably between the macro and micro levels of the system. Systems that engender emergence of some kind have a specific micro-macro feedback dynamic that becomes important to understanding their overall behavior. The dynamic where high-levels effect lower levels is a key part of emergence and it is called downward causation. Downward causation is the claim that in a system exhibiting emergence, the higher or macro-level as the locus of emergent phenomena exerts some kind of causal influence downwards on the lower level substrates in the system from which it originally emerged.

This interaction can be seen within biological organisms as the individual organs and tissues create the whole organism but then the organism, as a whole, feeds back to regulate the parts. Within societies individuals create institutions but then those institutions feedback to constrain the individuals towards the ends of the institution as a whole. Likewise, economies and financial markets involve a constant interplay between the macrostructure - such as market prices - and individual actions.

In all cases, emergence involves new descriptions of a system being used when we go from the micro-levels to the macro-level, this can be seen in many areas of science. For example, economics is divided into micro- and macro-economics with different terms used in the different domains. Likewise, theoretical physics remains divided between descriptions of physical systems on the micro-level in terms of quantum mechanic and on the macro-level in terms of general relativity. In the study of society, the micro and macro are divided into two different domains with their own lexicon; psychology for talking about the individual and sociology for talking about macro-level patterns within social systems. This is because phenomena, like social movements, only emerge out of the synchronized activities of many individuals and will thus not form part of the study of the individual but will only form part of the study of whole social groups.

The question that remains, though, is whether we form different descriptions on each level out of expediency and lack of knowledge, or do we do it because on a fundamental level it is not possible to fully derive an account of the macro-level from the micro-level? This difference is captured in the terms strong and weak emergence.

Weak emergence describes any emergent process that, given enough information, can theoretically be simulated by a computer. With the process of weak emergence, novel features and properties may emerge within a system that could not have been predicted a priori. However, once they have emerged it is possible, at least in theory, to derive them from underlying component parts; even if in practice this is often not computationally viable. Strongly emergent processes are ones that cannot be derived, even in theory, from a full understanding of the properties and interactions of the component parts on the lower level. The higher level emergent properties and features must then be understood as a whole in terms of the macro-level dynamics without referring to micro-level mechanics.

As an epistemology, emergence reflects a certain way of looking at the world that may be contrasted to reductionism. The idea of emergence can be identified as one of the very few central ideas behind systems thinking and the holistic paradigm. Both holism and reductionism lead to very different ways of seeing the world and fundamentally different approaches to conducting science.

The central question dividing the two paradigms is whether our world is expressed by just a few basic laws, which directly govern the most fundamental parts of nature, and through them, the more complex systems they compose? This would be the position held by reductionist approaches and leads naturally to a focus on studying ever more basic elementary parts. Or whether emergence is a fundamental part of our universe? In which case, it would not even theoretically be possible to reduce everything to an account involving only basic elementary part and thus we should focus our inquiry into understanding the abstract principles governing the patterns of organization that emerge on different levels. This would be the position held by more holistic approaches.

The basic reductionist intuition is that worldly events and processes are governed by just a few basic properties, linked through just a few basic physical laws, which results in what is called the completeness of physics. Everything that is, is physical, and every event, phenomena or process is caused by some low-level physical interaction.

The classical reductionist conception of the structure to the enterprise of science was provided by Oppenheim and Putnam in 1958, who made explicit a widespread view of the sciences as a hierarchical structure, with the domains of sciences higher up being shown to be composites constructed out of the entities of science lower down. This hierarchy can be thought of as a pyramid with reduction as the means for moving up and down its levels. At the bottom level we have fundamental physics, above these, we have other areas of physics, then upwards chemistry, biology, psychology, economics and other areas of the social sciences.

The result of this is a somewhat alienating view of the world with fundamental physical particles at the center and human activity as some form of peripheral activity, leading to a knowledge base that is very much removed from our everyday experience of the world. The idea of emergence leads to a very different conception of the enterprise of science, one that does not try to break things down but instead looks at the many different levels to find patterns and processes that are common to all. This approach is grounded in abstraction through identifying patterns that are common to all types of systems on all levels. Holism tries to understand the basic process of synthesis, synergies, and the dynamics of change that are seen throughout all kinds of systems. Instead of reducing things to a limited subset of physical elements and laws, it actively seeks diversity to derive the abstract patterns that emerge on all levels and are common to all systems. Here the unity of science is not looked for in elementary parts but in abstraction, abstract models that apply to all systems, social, physical, biological, engineered etc.

Patterns

A pattern defines some form of correlation between the states of elements within a system. All systems show some form of pattern, either in space between their parts, or over time within some process and these patterns can be understood as the product of some form of correlation between the constituent element's states. A correlation is a relationship of some kind between any two or more variables in which they change together over a period of time. A combination of correlations between elements forms a regular or intelligible pattern.

If there is no correlation between parts, then they are randomly associated. Randomness can be understood as the absence of organization and thus the opposite of a relational pattern. A random sequence of events, symbols or steps has no order and does not follow an intelligible pattern or combination.

Patterns have an underlying mathematical structure; indeed, mathematics in its modern form may be understood as the science of patterns. Similarly, in the sciences, theories explain and predict regularities in the world through modeling the correlated change in properties between things. In the everyday world, it is the aggregation of correlated phenomena into composite patterns that enables us to make sense of our environment, predict outcomes and act effectively within it.

A correlation is a mutual relationship or connection between two or more things, where the properties associated with each element in the relation change with respect to each other in some way. The term correlation derives from the Latin *cor*-'together' and *-relatio* 'relation'— as it refers to things that more or go together. A correlation can describe any related change in elements, change in the density of two materials, changes in the physiological form of two organism, changes in beliefs and values of a society over time etc. Science proceeds by making a series of empirical observations and then tries to draw relations between thing; "this thing is bigger than that," "these people are more powerful than those," "all of the springs for the past ten years has been warm" etc. Out of these observed empirical correlations we can develop patterns of organization that help to make the world more intelligible.

Three primary factors to the type of correlation are; whether the relationship is positive or negative, the strength of the relationship and whether it is linear or nonlinear. A positive correlation means the variables move in the same direction, both rising or falling together. For example, the amount of money one has in a bank account will be positively correlated with the total amount of interest you earn on that account. The more money you have in the account, the more interest that will be earned, the less you have the less will be earned. In a negative correlation, both variables move in the opposite direction, like the relationship between the amount of fuel in one's car and the distance that has been traveled; the fuel will go down as the distance traveled increases.

Correlations have varying degrees of strength. A strong correlation means the variables move together exactly and proportionally. A weak correlation means that the relationship between variables is only partial. For example, there is a correlation between age and health, but it is relatively weak. Given someone's age we can not predict their health, some young people are often ill while some senior citizens are very healthy. There is a general correlation but it is not direct and thus relatively weak.

This relation then connects the elements into some combined form of organization where a change within one element will - at least partially be - associated with the modification of another, thus creating a pattern that represents the combined organization. The term "correlation does not imply causation" highlights the fact that this connection does not have to be direct, the connection may be intermediated or generated by one or many other variables.

A linear correlation describes an association where the ratio of change between the variables stays constant over time - thus creating a straight line when plotted. A nonlinear correlation describes how the associated change in each may itself change over time, i.e. the proportionality to the change between elements can change over time, thus mapping out a graph that is not a straight line. For example, there is often a linear relationship between the distance one has to travel and how long it takes to get to the destination, if the destination is twice as far it will take twice as long to drive there. But there is a nonlinear correlation between the size of a factory and the cost of maintaining it, to operate a plant of one thousand square meters would not cost twice as much as that of five hundred square meters, because of the synergies to economies of scale.

The robustness of the pattern is then a function of the number of relations and the strength of the correlations within those relations. If all the parts are interconnected and change exactly with all the others, then we have a strong or robust pattern. An example of this would be an army troop marching together, every member's state is supposed to correlate directly to every other member, making for a very strong pattern that we would identify immediately. These strong linear correlations are much easier to predict because of their manifest, direct relations and proportionality. We could for example easily predict what one of the members of the army troop will do when the others move.

Likewise, the robustness of the pattern is low when there are few connections and a weak nonlinear correlation between them. For example, there might be a weak pattern between the price of rice in Thailand and the price of eggs in Norway. With these weaker nonlinear patterns, the correlations are not manifest - they may be intermediated by many different elements - and they may be nonlinear. We do not know all of the factors that might connect the price of rice in Thailand, and that of eggs in Norway and these connections will likely change with varying degrees over time.

Symmetry is probably the most fundamental organizing principle to patterns. Symmetry describes what is similar about two or more elements. Symmetry helps us to capture the fundamental concepts of "sameness" and difference. Symmetry - in the abstract - defines how two things are the same in some way, while asymmetry defines how they are different. Symmetry is a fundamental feature of pattern formation, a pervasive phenomenon in our world found in the spatial and geometric relations between forms as can be seen in architecture, in how events take place over time, in the composition of music or of a sculpture. Symmetry is at the heart of modern mathematics being studied in the area called group theory. Symmetry has become fundamental to our understanding of the basic laws of physics for almost a century now. This concept has become one of the most powerful tools in theoretical physics because it has become evident that practically all laws of nature originate in symmetries. Nobel laureate PW Anderson wrote in his 1972 article *More is Different* that "it is only slightly overstating the case to say that physics is the study of symmetry."

A symmetry - in the abstract sense - describes a rule that will map or transform one element in a relation to another. For example, a snowflake has a geometric symmetry to its form, what is called a reflection symmetry, where one side can be transformed into the other by applying a reflection transformation. In this way the original element has not changed, we have just applied some transformation to it to derive another related element if we took the transformation away we would return back to the original element. Symmetry is an important property of both physical and abstract systems, and it may be displayed in precise mathematical terms or in more aesthetic terms.

Asymmetry is the absence of, or a violation of, symmetry. Asymmetry may be understood as a lack of perceived transformation that will map one element in the relation to another. Asymmetry, in its generalized sense, describes how things are different within some frame of reference. For example if we take a tree that is asymmetrical, having more branches on one side than another, then unlike a symmetrical tree - where we could simply perform a flip operation on one side to get the other - with this asymmetric tree there is no transformation that will map one side to the other, thus it is

asymmetric and we would say that one side of the tree is different to the other because of this asymmetry.

Symmetry and asymmetry can be understood to be relative to frame of reference or information. As we go to higher levels of abstraction we see that things that before appeared different, i.e. without transformation to map between them, now on a higher level of abstraction have symmetry.

Symmetry can be used to define the level of order within a system. Order means the arrangement or disposition of people or things in relation to each other according to a particular sequence, pattern, or method. This particular sequence that defines order can be understood as some transformation or symmetry between the elements states or over time. If we look at an object like an isosceles triangle or a square they will appear much more orderly than an irregular triangle because the isosceles triangle and square have more symmetry to them. Symmetries help us to grasp the world around us and to find order in it by compressing information.

Patterns that are symmetric can be defined in terms of some subset of the entire pattern and a transformation that when performed will generate the other forms within the pattern. For example, if we had a number pattern of say 2, 4, 8, 16, 32 we would not need to itemize each element in the set we could just state the first element and then the transformation of doubling that would generate all the elements in the pattern after this. Thus we could generate the whole set with just one piece of data and one rule. Because of this symmetry within the pattern we can now represent or describe the whole pattern with only a very limited amount of information. The same goes for any ordered system. Because symmetries define order we can describe an ordered system in terms of some small set of data and transformations, in so doing compress the amount of information needed to describe it.

Inversely, because asymmetry in the generalized sense means a breaking of a rule, for every asymmetry we will need to add more information. If our pattern was 2, 4, 8, 16, 18, 36, 72 our original rule of doubling each time has now been broken and we would have to add an extra rule to account for this broken symmetry and thus more information to describe the system. The same is true for any broken symmetry, if a car had a big dent on one side of it - a broken symmetry - we would have to add an extra piece of information to describe it.

Whereas symmetry describes simple patterns, in that there is a small set of rules governing the difference between the parts that can be used to generate the whole pattern; once we understand those rules the pattern will appear relatively simple. Asymmetry describes complicated patterns in that it requires significantly more information and rules to generate the whole pattern. A tangled ball of string is complicated because there are no symmetries to the pattern, it would require a detailed description of the entire pattern to understand it fully. Complexity can be understood as some interaction between symmetry and asymmetry to create a pattern that has order but is also somewhat random and chaotic; it is this interplay between the two that is a defining feature to complexity patterns of all kind.

Synergies

A synergy is an interaction or coordination between two or more elements or organizations to produce a combined effect greater – or less – than the sum of their separate effects. A synergy is a particular type of interaction between parts; it is a nonlinear interaction where the specific way the parts interact creates an effect greater or less than the simple sum of their effects in isolation.

Synergistic interactions are pervasive in our world. Examples of synergies include ants working together in a colony to achieve result far greater than the sum of their activities in isolation, two drugs having a combined effect that is greater than each taken in isolation, or two companies merging to create a more efficient combined enterprise through the combination of their capabilities and resources

Synergies can be understood with reference to linear and nonlinear relations. A nonlinear relation can be understood as one that in some way adds or subtracts value above that of the components due to the specific way that they are combined or arranged.

Nonlinear relations can be best interpreted when contrasted with linear relations. A linear relationship is one where the properties of the elements do not change due to the connections. If we have two apples, and we create a relationship between them by placing them side by side on a table this connection will not affect the properties of each apple; the properties remain invariant. Because of this, the relationship does not change, add or subtract value from the combined organization. Linear relations define the connections and arrangement between objective properties of things – what is also called a primary quality of an entity. If an object's properties are objective, they do not change depending on the context or connection with other things. For example, the solidity, extension of, or the molecular motion within a substance are objective; they do not change depending on the context.

If we take two or more things with these objective properties when we combine them their properties will not change. Thus the combined organization will be simply a summation of the original part's properties. There are many examples of linear relations such as a group of unassociated people waiting at a train station, there is no particular relationship or organization between them; thus the overall organization is simply the summation of its parts. Likewise, a pile of bricks has no special relation between the parts and thus gain the collection is nothing more than a linear combination of the components in isolation.

A nonlinear relationship between things is one where the interaction between them adds or subtracts value from the overall combined organization. With a nonlinear relation the parts interact in a particular fashion to create a combined organization that is different in some way from the two parts taken in isolation. A nonlinear relation involves an interaction between the subjective properties of the parts. Unlike an objective property that is exhibited under all conditions, a subjective property is one that is only present under certain environmental condition – what may be called a secondary quality.

Color is an example of a subjective or secondary quality; objects do not have color as intrinsic objective properties. The sea and the sky are not innately blue in the way that we perceive them, nor are other things, they simply have atoms absorbing and emitting photons at particular wavelengths. To get the phenomenon of color we need not only the object but also the specific visual system and cognitive processing of humans or other creatures. Thus the color of an object is an interaction between the two; the object has the potential to induce the effective of color but it is only manifest when in relation to a creature with the appropriate photoreceptive system. The color of an object is subjective, it is not manifest in all contexts, it is only exhibited in particular contexts given the right interaction.

Another example would be sexual reproduction or love, each partner in the relation has the potential to produce another creature or to fall in love, but the potential is only actualized given the particular interaction with another member of its species. Likewise in a chemical reaction, certain substances have the potential to react violently, but only given the particular interaction with another specific chemical. With a synergy the component parts have some subjective potential and that potential is released in the interaction to add or subtract value from the combined organization. Synergistic relations involve a bidirectional, mutual interaction between the parts, the mutual effect of components or systems on each other. This relation can be thought of as feedback between the components as there is an influence that is reciprocal in nature. Which is in contrast to a linear relation that may only be unidirectional.

Nonlinear synergistic relations require specific subjective conditions or context. Linear relationships are more objective meaning they will hold under a wide range of circumstances. Linear relations are much more robust or flexible in that we can change things around and still get the same results. In contrast, with synergistic interactions results may change widely given small changes in the arrangement of the organization. These linear combinations have special features to them, what are called in mathematics, commutativity, associativity and distribution.

The commutative rule says we can swap things around and still get the same combined outcome. For example, with the people waiting for the train, it makes little difference who was next to who because they had no special relation between them. Now if we added specific synergistic relationships between some of the parts – say this was a group of school children who all had particular friends within the group that they like talking to – now it would matter how we arrange them. If they are all arranged in a specific manner so as to be beside their friends, then we would get a very different overall outcome – lots of people talking – than if they were arranged in another way so that they were not close to their friends.

The associative rule says that it does not matter how we group things. For example, if we had a set of unordered bricks we could group them into equal piles or have one pile with a lot and another with just a few bricks, it makes no difference to the overall combined organization because of the linear relations between them. However, organizations with nonlinear synergistic relations are not like this. A company, for example, has department groupings that enable it to work efficiently. If we recombined and regrouped the whole organization in a random fashion, the whole organization would be significantly affected as productivity would be greatly reduced.

Finally, the distributive rule, in a general sense, states that doing something – performing an operation – to the combined organization is the same as doing it to the different parts in isolation. Watering one's two seedbeds of tomatoes separately would have the same effect as watering them both together as one seedbed. However, if we were to do this with a system that had nonlinear synergistic relations, the same would not be true. For example telling each partner in a relationship separately that the other is cheating on them would not have the same effect as telling them both together.

Within linear systems, because the properties of the elements do not change depending on the relations and context we can use these rules to deal with complex phenomena by breaking them up. We can simplify things by decomposing them, treating the subsystems in isolation and then recombining them to get the same result as if we had dealt with the system as a whole. We can take for example a math problem and distribute it out into smaller parts and then add all the pieces back together to get an account of the whole. Likewise, we could, for example, paint our house by breaking the process up into stages. We can do this because there is no particular relationship between the activities; by performing each one in isolation, we can complete the whole.

Whenever there is some form of synergistic relation in the system we can not do this, because these "special" relations that add or subtract value to the organization may be lost when decomposing and rearranging the system. Because synergistic relations are only present under certain circumstances, if we break things up or move things around arbitrarily we lose them, and the outcome will not be the same as before we performed the operation.

With a linear relationship – where the interaction between the parts does not add or subtract value from the whole – the inputs and outputs to the system are proportional. Like hitting a ball with a bat, the ball moves off at a velocity that is directly related to the force that it was hit with. In contrast with nonlinear synergistic relations the input and output can be disproportional because the relationships between the parts can add or subtract significant value. For example, if we invested money in a company that had many positive synergies, where the team was working very well together, we may well get a hugely disproportionate return on our investment.

Positive & Negative Synergies

A synergy is a nonlinear relationship between two or more elements whereby they generate a combined outcome that is more or less than the sum of their parts taken separately, due to their capacity to work together or against each other. The idea of a synergy is one of the core concepts within system theory in that it forms the foundation to the idea of emergence and the concept of a system as being more than the sum of its parts.

Although the basic idea of a synergy is of importance in many different domains of science, management and engineering its existence as a generic term is not always noted. The term takes various forms in different areas, being strongly associated with such concepts as emergence, interdependence, cooperation, self-organization, order and interactions among others.

The interactions between the parts of an organization can be defined as either linear or nonlinear. Linear relations are those that simply combine or recombine the parts without that interaction changing the overall system. For example, a zero-sum game involves linear interactions where the parts exchange resources but the overall amount does not change. Thus linear interactions do not add or subtract value above that pertaining to the elements within the system. In contrast, a nonlinear interaction is one that adds or subtracts value from the whole. A non-zero sum game would be an example of this, due to the cooperation between actors the overall pie can get bigger, thus adding value to the combined organization.

Synergies define a nonlinear relation between the parts of an organization. Unlike linear interactions that are context-independent, synergies are context-dependent, the overall value is added out of the interaction between two or more specific parts. With synergies, it is the interaction that adds or subtracts the value and that interaction is dependent upon the specific components that are combined within it. For example, synergies will only occur within a business if the right people are connected in the right way, in the right context. We can not just connect any people in any fashion, doing so would remove the synergies.

Interdependence is a foundational part of synergies. The elements within a positive synergy are interdependent in that they have to each perform different functions or roles with respect to each other. For example, a football team works synergistically due to the various roles that the members occupy within the whole organization. Moreover, not only do they have different roles with the team they are also adapted to and interdependent on the specific characteristics of the other members of their team. If we suddenly switched the quarterback from one team and put them in another we would likely find the performance reduced substantially. In this way, synergies are always contingent upon the particular context within which the component parts have developed interdependencies with other elements within that system. This is in contrast to linear relations that do not engender interdependencies. For example, in building a brick wall we could just swap out one of the bricks and replace it with another of the same specification or move the block from one place in the wall to another.

This interdependence between the parts within a synergy is a function of the degree to which they are both integrated and differentiated. Differentiation means that the parts are performing different functions or occupying different states with respect to each other. All the players on a football team do different activities, the bees in a colony perform different functions, the cells, tissues, and organs of the human body perform different functional roles. This differentiation means that the parts can focus specifically on a limited number of functions and thus perform them more efficiently as they become more specialized.

But differentiation is of no use if those different parts can not be then reintegrated towards performing a collective function. All the members of a team or the members of a business need to

be coordinated into the common process required to achieve the overall objectives. Integration is just as important as differentiation in that it ensures all the different parts are working together. It is only by having the integration mechanism of market changes that we can all do different specialized occupations, without the macro systems for integrating all our differentiated skills we would not be able to specialize.

In every synergy and relation of interdependence there is a dynamic of differentiation between the parts - that is to say, they must be doing different things or in some way have different form and structure - and of integration so that they create a combined organization. The degree of effectiveness of the organization will be largely defined by the extent to which this is achieved and how balanced the dynamic between the two is.

A positive synergy describes how the combined organization is more than the sum of its parts due to the parts working together constructively. Indeed, the term synergy can be defined as a measure of the effectiveness of the joint efforts of various subsystems acting in coordination. Positive synergies are a product of the elements effectively achieving both differentiation and integration. For example, in the process of brainstorming to come up with new ideas, the process will be most successful if given a wide diversity of ideas and the capacity to synthesis those different ideas into a finished outcome. Positive synergies in group decisions may well include the generation of more ideas, more creative solutions, a greater acceptance of diversity, increased acceptance of the decision by team members and a greater capacity to work together towards delivering a finished solution.

The degree of positive synergy in a group can be understood as a function of the level of both differentiation and integration added to how balanced these two are. The more specialized the parts, the greater their integration and the greater the balance between these the more functional the overall system will be. The human body can perform the many functions that it can because of both the extremely high level of differentiation between its parts - creating interdependency - and its capacity to integrate those. Simpler organisms lack this high degree of differentiation and integration and thus lack many of the functions of the human body. Likewise the same would apply to a multinational corporation, to a production process, or a technology. A modern car delivers much greater functionality than one of fifty years ago; but to do this, a car today has about 30,000 parts, which are all well integrated. A computer can perform many more functions than a calculator because it has many more specialized subsystems that are coordinated. Thus it is this combination of differentiation and integration that creates positive synergies and creates the functionality that adds value to the whole organization.

A negative synergy is a nonlinear interaction where the combined outcome is less than the sum of the parts effects taken in isolation. A good example of a negative synergy is competition, such as an arms race, it is the specific way the two parties interact that we get the overall outcome that is counter-productive and detrimental to all. Another example of a negative synergy would be two creatures fighting over the same territory.

Negative synergies can be understood as the failure of the parts engendered in the relation to differentiate or integrate. Either the parts become too different without integrating - meaning they are doing very different things and can find no way to interoperate - or vice versa they become too integrated and similar. An example of the former might be a body of knowledge becoming too specialized without finding ways to interrelate the different domains, the result being fragmentation. The same could be true of a family as the members grow-up and become focused on their particular lives without being able to find common ground between them. Worse than fragmentation the parts can end up pulling in different counteractive directions, such as in a meeting where the members have different ideas while needing to find a common outcome thus leading to deadlock and potential conflict. Too much integration between the parts can likewise create a negative synergy as all the parts come to occupy the same state the full set of possibilities are not explored, there is a lack of specialization and diversity, and there can be crowding out as all the parts try to occupy the same state or function.

A good analogy for the negative synergies of both over integration and over differentiation would be the example of free market capitalism and communism. Free market capitalism often overemphasizes competition resulting in a lack of integration, ending up with millions of products on the market, all competing as they assert their difference and merits while many of them are the same, resulting in an excess of economic activity going into differentiation and a loss of overall productivity. Likewise, communism worked in the opposite direction with an over emphasis on the commonality between people and their economic activities, with the result being a lack of capacity to harness the individuals' diverse motives.

Pattern Formation

A pattern is any set of correlations between the states of elements within a system. Pattern formation refers to the process through which a coherent set of associations between elements states is formed and persists over some period. A primary question we are interested in when studying any pattern is the question of how was it generated or formed? In answering this question, we can make a fundamental distinction between those patterns that were created through order being imposed by some other external organization, or those that were created through the pattern being internally generated.

The term pattern formation typically refers to internally generated patterns. By pattern formation, we mean that certain systems have the ability to self-organize into structured states from initially unstructured or homogeneous states. This internal pattern formation process is a universal phenomenon as it has been identified in physical, biological, economic and social systems.

One primary characteristic of emergent pattern formation is the lack of centralized control. This process of emergent pattern formation uses only local dynamics to influence the systems global behavior. No single part of the organization coordinates the macro-level behavior. Thus it is typically not possible to directly control the macro level pattern; it is only possible to effect the micro level parts. Some examples of emergent patterns include the ripple patterns in a sand dune created by wind or water. Swarming and flocking are a well-known emergent behavior in many animal and insect communities. The development of traffic patterns like gridlock, as well as the formation and adoption of new cultures, are internally generated patterns formed out of local interactions between the parts.

A central characteristic of internally generated pattern formation is positive feedback. Positive feedback is a process whereby more begets more, i.e. the more we have of something the more we will get of it in the future. Compound interest is an example of positive feedback, the more money in an account, the more interest that will accrue, which will mean more money in the account in the future, which will feedback to generate more interest, etc.

Within pattern formation, positive feedback works to enable some small - possibly random - event to take hold within a system and get amplified into a global pattern. For example, we can see positive feedback at work in the formation of sand dunes. Wind flowing across the sand carries small grains of sand with it. Slight unevenness in the surface of the ground creates a greater likelihood of grains collecting at a high point, creating an accumulation that then increases the size of this small lump, which then feeds back to increase the likelihood of more grains collecting at this point. Over time a dune can form, but of book, the dune can not go on building up forever, as it gets higher the sides get steeper and it becomes easier for grains to slide off. This is the negative feedback that places constraints on the pattern's formation. The fully covered pattern of the dunes or wind ripples is the product of many of these dunes building up until they meet each other to create a complete pattern.

Another example is the formation of convection cells within heated water. Convection cells emerge that transport the heat to the cooler regions near the surface of the liquid where the heat is given off. While at the same time cooled liquid is pushed to the bottom to be reheated creating a circular motion in the water and the formation of a pattern through the internal interaction between the warm and cool water. Similar patterns of hexagons and stripes can be found in very different physical and biological systems. For instance, these striped patterns are observed on the skin of zebras and on human fingerprints.

Ant foraging is another example of pattern formation through positive feedback. There is no central organization within an ant colony; coordination is achieved through the exchange of chemical pheromones between the ants, directly ant to ant as a means of communication. When an ant finds a food source it will excrete a pheromone on its way back to the colony. Other ants then pick-up on this and follow the source, if they too find food they will also leave a pheromone trail. Thus a stronger scent will build up, feeding back to induce more ants to follow. This is again a form of positive feedback that has led to the formation of the global pattern out of local interactions.

Another example within social organizations would be the development of open source projects. Many open source projects are initiated, but only a few become large scale stable patterns of organization like Wikipedia or Wordpress; again this is a product of positive feedback. The more people that contribute to a project and the more people participating in using the system, the more promotion and exposure it will get and the more attractive it will become for others to join.

In these above examples, one can note how it was in some way the energy that was being inputted to the system that enabled the internal pattern to form. Whether the wind moving the sand grains, the heat moving the water, the food for the ants, or people's work and attention driving the social organization. Emergent pattern formation may be internally generated, but it typically requires some input of energy, and this can help us to understand why the process takes place.

Often the elements in the system organize themselves to better intercept and transform the energy source - either randomly or purposefully. With many physical systems, this is somewhat random. With the accumulation of sand dunes, it was the random difference in the surface that enabled some areas to harness the wind's force towards building up their structure. But in many biological systems, this process is less random and in social systems it can be entirely purposeful.

In these cases, there is a random arrangement of elements on one level, and there is some energy source on another level, with the parts being only capable of intercepting that energy source through forming a particular global pattern. Any set of components that can form that particular pattern will then be able to access the energy, and this will create a positive feedback. Whereby the organization can get more energy and redistribute more to its constituent elements. This means more elements will be attracted to that configuration which will lead to the interception of more energy, etc.

For example, we can understand evolution in this fashion. Many of the energy sources that creatures use are only accessible by many different parts coordinating their activities towards intercepting and transforming the resource. Examples would include photosynthesis and mammalian digestion. These activities take the coordination of millions of cells and tissues; in isolation, none of the parts would be able to intercept these energy sources.

This idea is also captured in the theory of symbiogenesis which propounds the idea that several essential organelles of eukaryote cells originated as a symbiosis between separate single-celled organisms. Different more rudimentary organizations combined to form more complex organisms that were then capable of differentiating their internal features towards achieving greater overall functionality and thus becoming potentially more prolific.

For example, some coral can intercept and process light through photosynthesis by ingesting an alga that then forms part of the coral's structure. The algae can be found in the tissues of the coral, producing food through photosynthesis which is then taken in by the coral and in return, the host coral gives the algae a habitat.

This process of pattern formation can then be understood as a form of adaptation, a process whereby the system adapts to its environment. That is to say that emergent pattern formation is a process that involves the interaction between the system and its environment, whereby the parts of the system organize themselves in response to changes in the environment. In so doing we get the formation of higher level patterns to intercept new resources.

For example, we can ask how did we get life from nonliving elements? From the perspective of physics, the primary difference between an inanimate composite of carbon atoms and a living system is that the latter can intercept and process energy within their environment; but to do this the parts have to be arranged in a particular fashion.

One formulation of this theory based on thermodynamic principles, posits that when a combination of atoms is driven by a source of external energy - such as the sun - and surrounded by a heated environment - such as the sea or atmosphere - the system will often slowly restructure itself so as to dissipate more energy over time. This may lead under certain conditions to inanimate matter over prolonged periods of time adoption key physical attribute associated with life. An MIT researcher, Jeremy England, who formalized a theory of this kind said: "You start with a random clump of atoms, and if you shine light on it for long enough, it should not be so surprising that you get a plant."

A simple illustration of pattern formation within a social system can help to clarify the basic constituents of this process. Take ten beekeepers that all need to purchase sugar to feed their bees, but they each have a limited budget. If they each purchased in isolation, they would have to buy in individual packets from the local shop which would cost them too much to make it viable. If however, they formed some organization, they would be able to go to the distributor and purchase in bulk for the entire group at a discounted rate, making it a viable option.

We can see in this illustration how under the initial condition where there was a lack of organization - i.e. pattern - they were not able to intercept the resources. But by forming some pattern of organization - i.e. communicate a plan, collect all the money, purchase the sugar and then distribute it; all of which would require significant organization - they were able to intercept the resources that made the overall organization more viable and cable of developing. This same process is pervasive throughout socio-economic systems, as social systems are not dependent upon randomness in this process of pattern formation, but with the cognitive capabilities of the individuals can identify and purposefully work towards higher levels of organization.

Strong & Weak Emergence

Emergence describes how new higher level properties, patterns and functionality form as we put component parts and systems together. Emergence is a pervasive phenomenon in our world exhibited by virtually all types of complex entities, such as plants, animals, human societies, cultures, and economies. It is apparent that new levels of organization are formed as we put things together. When we look at a flower, we do not see a composition of molecules, cells, tissues and organs but in fact a whole flower, a corporation is given characteristics and legal rights that are not associated with any of its members. Although this emergence of patterns on different levels is apparent, what is not apparent is whether these emergent organizations are fully determined and understandable with reference to their basic component parts. Or do they exist in some way independently from the elemental that constitutes them; which would mean that they can not be entirely reducible to causal accounts derive from the elemental parts.

This distinction within emergence is captured in the terms strong and weak emergence. Where weak emergence refers to the idea that even though emergent phenomena are unexpected given the principles governing the low-level domain, they are still fully explicable only with reference to lower level phenomena. In contrast a high-level phenomenon is strongly emergent when it arises from the low-level domain, but facts concerning that event are not derivable even in theory from the features of the low-level domain.

Weak emergence is the phenomenon whereby new and unexpected patterns emerge due to the interaction between the parts, the sheer number of parts and interactions makes it computationally extremely difficult to compute based upon elementary parts and their rules. However, from the weak emergence perspective, given sufficient computational capability, it would be possible to derive the higher level phenomena and thus they are seen to be - theoretically - a derivative. New phenomena and patterns do emerge, but the reason we can not predict them is that the number of interactions between components of the system increases exponentially with the number of elements and we typically do not have the computation capabilities to deal with this.

With weak emergence, it is possible to compute the high-level phenomena but typically much easier just to look at it directly. Thus this weak emergence can be understood as a kind of explanatory emergence. That is to say, the emergent features are ontologically and causally derivative but in practice, they are explanatorily irreducible due to computational complexity. New rules may appear to emerge at the different levels but ultimately if we had the computational capabilities we would be able to understand all of the rules at the different levels with respect to the lower level rules. Thus if you had an extremely high level of computation capability you would not need to focus on the higher level phenomena but understand it from first principles; one could look at it as caused by purely discrete cellular parts.

A good example of weak emergence is a cellular automaton computer program like The Game of Life. The Game of Life is played on a grid of checkers where a cell can be either on or off; there are four simple rules as to whether a cell should be on or off depending on the state of it immediately surrounding neighbor cells. These simple rules, when computed, can create very complex and subtle emergent patterns that appear to have their own internal structure. Such as blinkers where a group of cells "blink" on and off or gliders that seem to glide across the screen all of which are emergent phenomena. The program exhibit sensitivity to initial conditions and it is very difficult to predict what will emerge based on the initial conditions and ground rules. Although these programs can create emergent patterns they are said to be weakly emergent because they are determined by the elementary rules, the starting state and because there is no downward causation; the macro level system does not change the micro level rules.

This weak emergence is characterized by the interaction between parts as the system evolves leading to computational complexity and the appearance of something new emerging, when in fact, it is theoretically reducible to causal accounts of the elementary parts. One can not in any straightforward way derive the high-level phenomena from the fundamental rules alone. Thus compact representation - such as equations - do not tell us very much of what is going on because we need to compute the interactions to produce the high-level phenomenon.

With weak emergence higher level phenomena do not affect the lower levels i.e. there is only upward causation present, the macro level is determined by the micro, but not vice versa. There is an asymmetrical flow of determination, macro level patterns are not doing anything over and above what the micro level events are doing to affect the positions and behavior of the elementary parts.

An event is thought to be strongly emergent when the high-level phenomenon derives from low-level events, but a complete description of the emergent pattern is not reducible, even in principle, to an account of the elementary parts and their interactions. Along with irreducibility, downward causality is commonly cited as a criterion for strong emergence.

Strong emergence entails the idea that something truly new emerges at the different levels of organization that can not theoretically be reducible to accounts of the elementary parts. The whole is something truly other than the parts. Thus it makes sense to talk about qualitatively different levels or dimensions to the system as the rules that apply on one level become replaced - at least partially - by rules of a qualitatively different nature on another level. These higher level patterns then can exert a downward cause on their constituent parts affecting their structure and functioning. Strong emergence describes the direct causal action of a high-level system upon its components; qualities produced this way are irreducible to the system's constituent parts.

One of the classical examples of strong emergence given is quantum entanglement. Quantum entanglement is a phenomenon within quantum physics where two particles spin states become "entangled" meaning the state of one is entirely dependent on the state of another. It has been empirically proven that the combined "entangled" organization determines the spin direction of the parts. The two particles can be lightyears away from each other but if the spin is changed on one this will be immediately reflected in a change in spin in the other. Thus the combined organization is in some way affecting a downward causation on the parts.

Another example from physics of strong emergence is water, being apparently unpredictable even given a meticulous analysis to the properties of its constituent atoms. It would appear that no computational description of the system can exist, for such a simulation would itself constitute a reduction of the system to its constituent parts. The emergent phenomenon, in this case, can not be described with reference only to fundamental rules but requires some form of macro level rule.

Whereas with closed systems the whole should be theoretically derived and caused by the parts, this, however, may not be the case in open systems. The parts form the whole but then the system has to interact with its environment as an entire system - not as a set of parts. This interaction requires that it perform particular functions and activities as a whole, such as walking which can only be achieved by a combination of two legs being coordinated and interdependent. For the system to operate within the environment as a whole it has to exert a downward effect on the parts in order to coordinate them towards performing macro-level processes that are required to interact and respond to the environment.

For example, we can derive the internal workings of a truck from the basic laws of engineering and physics. But using those same elementary internal rules we would never be able to derive why it is designed to drive on the left or right-hand side of the road. This phenomenon is not a product of the internal logic of the truck's design or of the physical laws that govern its workings. It is instead a product of the system's interaction with other systems and some historical, political and economic contingent within the environment; which may be seen to be exerting downward causation on the design of the truck.

The biologist Peter Corning illustrates this when he writes: "the debate about whether or not the whole can be predicted from the properties of the parts misses the point. Wholes produce unique combined effects, but many of these effects may be co-determined by the context and the interactions between the whole and its environment."

This question of strong and weak emergence is of major relevance as it goes a long way to defining whether we should focus our inquiry on the micro level structure and rules that give rise to the high-level phenomena; as would be the case if we posit that the world is weakly emergent. Or whether we should instead focus on the internal patterns of the emergent phenomenon, in their own right - as would follow naturally from a belief in strong emergence.

As such the concepts of strong and weak emergence form part of the foundations within the different scientific paradigms of reductionism and holism. Reductionism is based on the premise that complex phenomena can be broken down into simple "building blocks" from which high-level events can be reconstructed. Thus a reductionist approach would typically ascribe to a weak emergent view of the world. Where complex macro level phenomena would admittedly take vast amounts of computation to derive from the basic physical building blocks but, given such a capability they would be fully explicable from the elementary parts and rules. The weak emergent theory inspires the idea that the goal of science is to understand the basic building blocks and rules for their combination and drives a quest for the "theory of everything" as seen to be found in some elementary particles; as is currently the quest of such approaches as string theory.

Systems thinking is instead interested in patterns and processes, it refers not to building blocks but more to patterns of organization and processes of change that are common to all types of systems on all scales without interest in reducing higher level phenomena to those of a lower level. As such the systems approach is built on a strong emergent view of the world. Given strong emergence, a "theory of everything" derived from elementary particles - such as strings - would end up being just one of many components necessary for a complete understanding of the universe and thus not necessarily the only one.

Within the systems paradigm, as all phenomena can not be simply reduced to an account of elementary parts, the goal of providing a unified description of the world is instead looked for in abstraction. Systems thinking looks at how these emergent patterns on different levels have similar dynamics and from this tries to develop abstract, generic models that are relevant to all scales because they capture the features inherent to emergent processes on all level.

Autopoiesis

The term autopoiesis derives from the word auto- meaning "self" and poiesis, meaning "creation or production." The term is used to describe a system's capacity to regenerate or recreate itself. A system may be considered autopoietic if the parts to the organization interact with each other in such a manner that they are continuously producing and maintaining the pattern and elementary parts that constitute the system. An autopoietic system is one that is organized to continuously reproduce its own parts and structure.

The term autopoiesis was introduced to systems theory by Citing Maturana and Varela, who defined an autopoietic system as "a closed topological space that 'continuously generates and specifies its own organization through its operation as a system of production of its components, and does this in an endless turnover of components.'"

Classical examples of autopoietic systems are biological organisms. The human body, for example, is believed to replenish every cell within itself over the book of a seven to ten year period. Likewise, the entire set of macromolecular elements within a given cell is regenerated approximately 100,000 times during its lifetime. During this extraordinary turnover of matter, the cell maintains its overall structure, coherence, and relative autonomy.

Of central interest within autopoiesis is the idea of how the parts create the pattern of organization but then this pattern of organization feedbacks to recreate the parts, this is even though the parts are creating the pattern of organization. The pattern appears to exist independently from any of the parts and can replace them without itself changing.

The cell maintains its identity and characteristics even though it incorporates at least a billion different constitutive elements during its lifecycle. This maintenance of integrity while the components themselves are being continuously or periodically decomposed and recomposed - created and destroyed, produced and consumed - forms part of the essence of autopoiesis.

On a grander scale, the global biosphere is another example of an autopoietic system. Persistent macro-level patterns, such as whole biomes or glaciers are continuously regenerating themselves to create a stable macro level pattern. This is captured in the Gaia hypothesis that states that the global ecosystem manages to maintain itself within the right parameters for confining life. The investigation of the Gaia hypothesis looks at how the global biosphere and the development of its constituent biotic and abiotic elements contribute to the stability of the global gas mix in the atmosphere, temperature, ocean salinity and acidity, and other factors to enable a homeostatic condition conducive to the perpetuation of life.

Many examples of this autopoietic process whereby macro level patterns recreate themselves can be seen in socio-cultural systems. Social institutions are enduring patterns of organization within a society that are built out of the coordinated choices of the individual members. Institutions like corporations, governments or democracy are recreated and perpetuated in the choices people make every day, to go to work, to believe in the value of a monetary currency, to vote, etc. Through society educating its children it recreates and continues the pattern of knowledge and culture of that society; as generations of people come and go the basic cultural pattern can remain somewhat invariant.

Parents, and society as a whole exhibit a strong drive to recreate their own inherited culture within their children, this appears a universe feature to human societies. More than this, people actively work to spread cultural patterns. For example, the Abrahamic religions, from their inception have shown a strong drive to "spread the word," to continue and expand the cultural pattern. This socio-cultural process of autopoiesis is mediated through communication. As the sociologist, Niklas

Luhmann noted: "Social systems use communications as their particular mode of autopoietic reproduction. Their elements are communication which are recursively produced and reproduced by a network of communications and which cannot exist outside such a network." In another writing Luhmann points out: "Elements are elements only for the system that employs them as units and they are such only through this system. This is formulated in the concept of autopoiesis."

Autopoiesis may be contrasted with allopoiesis which refers to a system creating some other external entity. Citing Maturana and Varela created a distinction between Autopoietic and allopoietic systems when they wrote: "Autopoietic systems are thus distinguished from allopoietic systems, which are Cartesian and which have as the product of their functioning something different from themselves".

The process whereby one system imposes a pattern on another external to itself may be called allopoiesis. For example, almost all engineered systems are produced through a human being imposing a preconceived pattern on a set of elements to create something other than themselves. A tractor factory would be a good example of this, the system is designed to produce tractors that are then externalized from the system. In contrary, some new 3D printers are specifically designed to be able to reproduce the parts needed to build another printer, a form of autopoiesis.

Integrative levels

An integrative level is a pattern of organization emerging on pre-existing phenomena of a lower level. Typical examples include life emerging on non-living substances, and consciousness emerging out of the nervous system. As components combine to produce larger functional wholes in hierarchical series, new properties emerge, and one cannot explain all of the properties at one level from an understanding of the components at the level below.

In Science paper by Alex Novikoff entitled "The concept of integrative levels and biology", he summarizes the theory as such: "The concept of integrative levels of organization is a general description of the evolution of matter through successive and higher orders of complexity and integration. It views the development of matter, from the cosmological changes resulting in the formation of the earth to the social changes in society, as continuous because it is never ending, and as discontinuous because it passes through a series of different levels of organization-physical, chemical, biological and sociological. In the continual evolution of matter, new levels of complexity are superimposed on the individual units by the organization and integration of those units into a single system, what were wholes on one level become parts on a higher one. Each level of organization possesses unique properties of structure and behavior which, though depending on the properties of the constituent elements appear only when these elements are combined in the new system. Knowledge of the laws of the lower level is necessary for a full understanding of the higher level; yet the unique properties of phenomena at the higher level can not be predicted, a priori, from the laws of the lower level. The laws describing the unique properties of each level are qualitatively distinct, and their discovery requires methods of research and analysis appropriate to the particular level. The laws express the new organizing relationship, i.e., the reciprocal relationships of elementary units to each other and to the unit system as a whole. The concept of integrative levels recognizes as equally essential for the purpose of scientific analysis both the isolation of the parts of a whole and their integration into the store of the whole."

Emergence addresses the organization of systems that self-organizes into layers of emergent wholes, systems that function according to nonreducible properties. This means that higher order patterns of a whole functional system, such as an ecosystem or whole societies, cannot be predicted or understood by a simple summation of the parts. These new properties emerge because the components interact, not because the basic nature of the components is changed. Water is more appropriately studied as a whole due to its emergent properties, but in the case of two cups on a table the lack of positive synergy precludes emergence; the system is thus best understood by simply analyzing each of the parts in isolation. To get good answers we must first ask the right questions. Quite different tools are needed for different levels; we do not use a microscope to study an entire ocean, a whole city, or the behavior of carbon dioxide in the atmosphere. Because of emergence our progress in solving environmental problems can be significantly slowed when the wrong level is focused upon and thus the wrong questions asked.

Principles

The theory of integrative levels puts forward a number of principles governing the structure of this hierarchical organization. Firstly, the higher levels depend on the lower levels, which would seem to be quite apparent. Higher level phenomena emerge from more basic constitutions, and without those constituent parts any form of emergence becomes impossible. This would imply that higher levels in the hierarchy are inherently more precarious due to this dependence on the lower levels. Secondly, the higher up the level the fewer the instances. This again would appear self-evident, as emergence involves a process of synthesis, or merging of different things. On every higher level,

the number of elements is thus reduced. Thirdly, the sequence of levels is often described as one of increasing complexity. This is far less self-evident and a somewhat debatable proposition, as it depends on one's definition of complexity, for which there are a number of different interpretations. One interpretation of complexity that would be congruent with this proposition is the idea that complexity is a combination of both integration and differentiation, which describes a system that has many diverse parts, which in turn are interconnected and interdependent. This is one interpretation of a complex system that is congruent with this hypothesis, because through the process of synthesis, systems are conceived that have increasingly many which are increasingly interdependent as well, thus enhancing complexity.

Hierarchy and emergence give rise to the design principle of encapsulation, describing how smaller subsystems are nested or encapsulated within larger structures. Key to this design pattern is the use of abstraction, meaning the successive removal of detail, as smaller locally specific subcomponents are nested within larger more generic processes and structures. Hierarchical encapsulation through abstraction is central to the structural design of complex systems of all kind and can be seen as a fundamental pattern of functioning ordered systems of complexity, as it distributes components out across different levels.

The so-called Parable of the Watchmakers illustrates this: There once were two watchmakers, named Hora and Tempus, who made very fine watches. The phones in their workshops rang frequently; new customers were constantly calling them. However, Hora prospered while Tempus became poorer and poorer. In the end, Tempus lost his shop. What was the reason behind this? The watches consisted of about 1000 parts each. The watches that Tempus made were designed such that, when he had to put down a partly assembled watch (for instance, to answer the phone), it immediately fell into pieces and had to be reassembled from the basic elements. Hora had designed his watches so that he could put together sub-assemblies of about ten components each. Ten of these sub-assemblies could be put together to make a larger sub-assembly. Finally, ten of the larger sub-assemblies constituted the whole watch. Each sub-assembly could be put down without falling apart. This is an example of abstraction and encapsulation within design that is central to dealing with complexity.

This hierarchical nested structure is seen within fractal forms that exhibit self-similarity across various scale of magnitude. Examples within ecosystems include everything from proteins and DNA to the capillaries in mammals, tree canopies, river networks and mountain ranges. This self-similarity or fractality implies a particular kind of structural composition or dynamic behaviour, wherein the fundamental features of the system exhibit an invariant, hierarchical organization that holds over a wide range of spatial scales and can be derived from simple iterative rules.

This process begets the emergence of a new level of organization: a structure that has some integrity to its parts or within which some process takes place, which creates its own distinct pattern. This then feeds back to shape and constrains the components on the local level. We get the emergence of some process, but for the process to take place there needs to be an enabling structure. That structure then defines some order to the system and the components must differentiate their states in order to perform the various structural and functional roles required to process the resource on the macro level. This is most evident from the way the human body as a whole regulates every local component of itself in order to enable global processes, such as respiration and digestion, to take place effectively. This is a micro to macro to micro feedback loop through the many different emergent levels from the cell to tissues to organs and the entire organism.

Micro-Macro Dynamic

The micro-macro dynamic describes a complex interaction between the micro and macro levels to a system. In every system there will be a degree of interdependence between the elements - by the standard definition of a system being a set of interdependent parts - this creates a dynamic where individuals elements that form autonomous components on one level are also part of some kind of a whole on another level; due to these interdependencies. The lower level, composed of the elementary parts is called the micro level and the higher level of the whole system is called the macro level. The micro-macro dynamic then describes how these two levels interact and coexist.

For example, within an economy, there is a micro level consisting of the individual agents and business organizations, and a macro level of the whole economy within which they all are interdependent. The actors may compete on the micro level, creating one type of dynamic - the market system - but for all to achieve optimal outcomes they also have to cooperate on the macro level - typically through government services and regulation - this creates two different levels, with two different sets of rules; those of the market and those of the public institutions, but both market and government have to interact and coexist.

This multi-dimensional nature to a system that results in the micro-macro dynamic is a product of synthesis and emergence. In many instances when we put elements together they do not simply remain discrete separate entities but they interact, co-evolve and they differentiate their states and function with respect to each other to become an interdependent whole, which comes to have properties and features that none of its parts possess. A whole new level of organization emergence that is different from the parts. This is made manifest in ecosystems; as they have co-evolved over millennia the parts are intricately interdependent forming a whole system that has features and dynamics independent from any of its parts and thus a two-tier system and a resulting emergent micro-macro dynamic. The whole ecosystem goes through processes of change - such as ecological succession - that are not associated with any of the parts but condition what creatures can viably exist within that macro regime.

When looking at some system - that exhibits any degree of complexity - we can identify two fundamentally different dynamics driving its behavior. Firstly it is not possible for something to exist in the world without it having physical parts; it is necessary to have physical parts to enable the whole, the physical parts are how the system does what it does, without them it would be incapable of acting or affecting its environment. Thus the lower levels of the hierarchy are subject to physical constraints, the limits of possibility come from lower levels in the hierarchy.

But equally, the system is conditioned by the whole. All systems exist in some environment and that environment typically acts on the system as a whole. The environment places constraints on the whole which are then exerted downward to the parts. For example, a wheel, if placed on an incline will roll down the slope. The influence is coming from the environment that is making the system as a whole roll down the incline which is conditioning the state and position of the individual molecular parts of the object.

The lower levels answer the question "How?" they are the building blocks that make the system physically possible and without which the system could not exist. Although there will be many possibilities only a limited number of these possibilities will be relevant to the context within which the macro level exists. Physically there are trillions of possible permutations of biological organisms but only those that "fit" with their environment will persist. As the ultimate reason why we have some permutations and not others is because of the context within which they exist - the macro level - it is the upper, macro-level that answers the question "Why?"

Timothy F. Allen, Professor of Botany at the University of Wisconsin-Madison illustrates this as such: "Mammals have five digits. There is no physical reason for mammals having five digits on their hands and feet, because it comes not from physical limits, but from the constraints of having a mammal heritage. Any number of the digits is possible within the physical limits, but in mammals only five digits are allowed by the biological constraints. Constraints come from above, while the limits as to what is possible come from below. The concept of hierarchy becomes confused unless one makes the distinction between limits from below and limits from above. The distinction between mechanisms below and purposes above turn on the issue of constraint versus possibility. Forget the distinction, and biology becomes pointlessly confused, impossibly complicated chemistry, while chemistry becomes unwieldy physics."

Each level in the hierarchy operates under different rules, is driven by different objectives, or shaped by different factors. The macro level is responsible for maintaining balance, stability and functionality within the whole system and is responsible for the system interacting with its environment; all of these factors place certain conditions on the operation of the macro-level. The macro system - if teleological - has to understand the environment, the possibilities within that environment, select the most appropriate one and then organize the parts towards that direction, in so doing it defines the question of "Why?" i.e. the higher meaning to why the parts are doing what they are doing. The parts have a higher meaning to their operation when they are operating in enabling this overall outcome. The constraints as determined by the environment come from above.

The macro-level is primarily concerned with the integrity of the whole system and ensuring that the micro-level can be effectively coordinated. Within social systems, the macro level may often be seen to be conservative in nature, departure from the established rules that ensure the functioning and perpetuation of the whole are seen as deviant. A good example of this would be religions, that are traditionally designed as the macro-level integrative structure to a culture and society. Religions provide a description of the world that is integrated into one ultimate conception of reality, God. Religions provide people with a situation in that reality, the right path to follow and a moral code as to how to act and conduct one's life so as to follow that perceived correct path.

Religions see the departure of the individual from that ultimate context - such as acting purely in one's own self-interest - as a form of deviant behavior. These faith systems - and spirituality of all form - are designed to provide meaning to people's life, the ultimate answers to the question "Why?" by situating them within and aligning them with some larger environmental context, within which they are believed to play a significant part and thus gain significance, meaning and a source for answering why they act in a certain way.

Another example would be socialist political regimes, that are holistic in their ideals; purportedly giving precedence to the whole of society over any of its constituent subsystems or members. Again socialist ideologies are designed to give coherence and integrity to the whole social system. The emphasis is on equality between member parts and their contribution towards the whole organization, individual identity, agendas, and motives are expected to be subsumed by the whole. In forming part of the whole members find their identity, it is the macro level set of rules that give context and meaning to their lives and the preservation of that integrity is seen as paramount. This is a defining factor to the macro-level of organization; a set of rules that are in place to preserve the integration of the entire organization.

The micro-level to a system is constituted by - and driven by - discrete component parts. Unlike the macro level elements can be quantified as discrete and autonomous components, typically having a material instantiation. This capacity to defined discrete and autonomous parts is a defining feature of the rules governing the micro level, in that it enables us to ascribe objective properties and attributes to specific parts.

When we take something as a whole we can not compare it to other things, and ascribe attributes to it based upon comparisons, to do so would be to reduce it to a part within some larger system. But parts of a system can always have properties and we can in some way measure those

attributes by comparing the parts, one person is small because another is larger, one car is efficient because another is inefficient - components by the very definition of there being more than one of them - always have properties in relation to other things.

By creating a fixed set of differences between things we can create a metric system; making it possible to measure things. Whereas the macro-level becomes defined by the integrity of the network between the parts and the integrity of whole processes, the micro level is defined more by the static properties of discrete entities and the metric system created by the difference between them.

Within social systems, the micro-level is governed by the individual agents and their local agendas. Free market economies are good examples of systems that are predominantly based on micro-level rules. The dynamics of the system are - at least theoretically - driven by the actions and motives of the individual agents without reference to the macro-level; the system is seen to be nothing more than the micro-level components and interactions governed by supply and demand constraints. Here we can note that the external environment is not considered; free market economic models see the natural environment as simply a commodity source within the economy. These models do not recognize the economy as forming part of some larger environment - social or natural - with that larger environment placing any macro-level constraints on the system. Free market economies are focused on discrete tangible products and agents acting as individuals without macro-level constraints.

Liberalism - or at least classical Liberalism - would be another example of a micro-level, component based socio-political paradigm, as individual liberty, freedom of expression, diversity of opinions etc. are given precedence over macro-level socio-political integration. Liberalism leads to the decoupling of socio-political institutions from the overarching macro-level structures of religious institutions. With the rise of liberalism in the seventeenth and eighteenth-century traditional institutions that we designed to maintain overall socio-cultural coherence have often been replaced by a new set of institutions driven by individual choice, and freedom for people to follow their own independent agendas.

Whereas the micro level to a system can be understood through analytical methods, taking things apart to examine them in isolation, the macro-level typically can not, as understanding whole processes require synthetic reasoning; the putting of things together to see what emerges.

When we put things together we combine their attributes, thus as we go up the hierarchy we have fewer and fewer properties as they become more generic and abstract, relating to the whole. For example, there are many things that we can not say about our universe when taken as a whole, we can not say it is heavy or light, big or small. But if there was another universe then we could ascribe these properties to it based upon its relation to the other element of its kind.

Inversely through analysis as we break things down we decompose a whole that had generic properties into parts that have properties in relation to each other; thus we can then create new categories out of these parts. For example, the neutron in the center of an atom when taken as a whole has no property of spin, but when we decompose a neutron we can talk about the spin of the quarks in relation to each other. In analyzing the body and taking it apart we would find that the different parts to its anatomy have many more properties than the whole.

A central question within the micro-macro dynamic is that of integration between the two different levels. Two distinctly different levels emerge with different rules governing each, but they then need to be reconciled in some way to create an overall functioning system. Physical, chemical, biological, economic, social and cultural systems all exhibit this micro-macro dynamic and how the system comes to reconcile this dynamic will form a primary determinate in its identity and overall structure. When the system fails to coordinate its parts into a functioning whole then it will become dominated by the interaction of its parts on the micro-level; what may be called a component based regime. An example of this might be the Warring States period in Chinese history, when the region of the Zhou Dynasty was divided between 8 states, during this period there was no overall organization but the state of Chinese civilization was defined by these component parts and their interaction.

Inversely the system may resolve the dynamic by forming a macro-level structure that constraints the micro level and thus again reduces the system to one level; removing the complex interaction between the two levels. Continuing with the example of Chinese civilization the communist political system of the past sixty years or so has been a form of social system where the macro-level regime has supervened over the micro-level.

Both of these dynamics create a single basin of attraction, but more complex evolutionary systems exist at the interaction between the two regimes. Systems that evolve engender a more complex dynamic between the micro and macro levels, with the micro level generating new possibilities and the macro level placing the conditions that select from this variety as the system develops over time. Natural selection is, of book, the classical example of this but equally, a functioning democracy would be another example.

Edge of Chaos

The term edge of chaos is used to denote a transition space between order and disorder that is hypothesized to exist within a wide variety of systems. This transition zone between the two regimes is known as the edge of chaos, a region of bounded instability that engenders a constant dynamic interplay between order and disorder. This point or interface between the two is hypothesized to be a locus for maximum complexity and the dynamics driving evolution.

Mitchell Waldrop in his book *Complexity: The Emerging Science at the Edge of Order and Chaos* describes the term as such: "Right between the two extremes [order and chaos] ... at a kind of abstract phase transition called the edge of chaos, you also find complexity: a class of behaviours in which the components of the system never quite lock into place, yet never dissolve into turbulence, either. These are the systems that are both stable enough to store information, and yet evanescent enough to transmit it. These are systems that can be organized to perform complex computations, to react to the world, to be spontaneous, adaptive, and alive."

Much of the ideas surrounding the edge of chaos hypothesis derive from chaos theory. The idea of order and chaos has fascinated people from many different domains for millennia, but these very big ideas in their abstract form have remained largely outside the scope of modern science. With the rise of chaos theory and complexity, a new language formed for approaching these fundamental concepts. Chaos theory has worked to explore the transition between order and disorder, which often occurs in surprising ways.

Chaos theory has provided some understanding of how systems turn chaotic, while the study of synchronization has dealt with the question of how things come into and go out of coordination. Another central element of this enterprise has been the growing understanding of the process of self-organization. With the theory of self-organized criticality and catastrophe theory, we are starting to get real quantitative models as to how these macro-level processes of change between order and chaos might develop along the edge of chaos.

Since the beginnings of the modern era, science has largely dealt with change as moving from one stable equilibrium to another. The Newtonian paradigm did not cope well with the random, near-chaotic messiness of the actual transition itself. Much of physics, chemistry, and other fields, have been focused on the study of equilibria. Engineers and economists similarly favored equilibrium conditions because neither analysis nor modeling techniques available to them could handle these in-between transition states. However, inexpensive computational power has changed this. Nonequilibrium and nonlinear simulations are now possible. These developments, along with the study of complex systems, have enabled us to better understand the dynamics of this "in-betweenness" or messy state at the edge of chaos that is a lot more representative of how our world actually looks in reality.

One of the original stimuli that lead to the idea of the edge of chaos come from computer experiments with cellular automata done by Christopher Langton. Christopher Langton defined a quantity called λ for any cellular automata. Lower values of λ corresponded to rulesets with less change. Higher λ s led to more change. He showed that cellular automata with low λ s were more prone to rapidly moving towards a balanced or static point. Those with a high λ value tended toward complete randomness. A λ value toward the mid-range, a "critical" λ , resulted in programs that could generate long periods of complex aperiodic non-random behavior before settling into either a fixed point or randomness. This was the edge of chaos. Kauffman saw in these programs new and useful developments as emerging "on the edge of chaos," the boundary between ordered and chaotic regimes.

In the paper Kauffman published on the topic he wrote: "Above a certain level of 'complexity', the process of synthesis is also degenerative. In other words, we find that there exist an upper limit as well as a lower limit on the "complexity" of a system if the process of synthesis is to be non-degenerative, constructive, or open-ended. We also find that these upper and lower bounds seem to be fairly close together and are located in the vicinity of a phase transition. As the systems near the phase transition exhibit a range of behaviors which reflects the phenomenology of computations surprisingly well, we suggest that we can locate computation within the spectrum of dynamical behaviors at a phase transition here at the "edge of chaos".

This edge of chaos condition within cellular automata was previously noted by Von Neumann and it can be seen within Conway's game of life where he had to figure out how to get rules that would create complex patterns. If the rules to the game of life are slightly changed they will not produce interesting phenomena.

These ideas originating in computer programs have since been generalized to all forms of systems that exhibit complex evolutionary behavior. Today, in the sciences in general, the phrase "edge of chaos" has come to refer to a metaphor that some physical, biological, economic and social systems operate in a region between order and either complete randomness or chaos, where the complexity is maximal. This edge of chaos phenomenon is thought to be a characteristic of many different types of complex systems, as complexity can not be understood in terms of simple symmetries but neither is random: it is some combination of both. This in-between state that complexity is thought to lie in, makes it un-amenable to our traditional scientific methods.

In the book Complexity and Organization, the authors write: "Nothing novel can emerge from systems with high degrees of order and stability - for example, crystals, incestuous communities, or regulated industries. On the other hand, complete chaotic systems, such as stampedes, riots, rage, or the early year of the French Revolution, are too formless to coalesce. Generative complexity takes place in the boundary between regality and randomness."

A system with no order can not exhibit useful behavior. But also a system with too much order can become over constrained and likewise not exhibit functional results. It is possible that processes organize themselves into conditions so complex that no usable functionality can result from it i.e. there can be too much historical memory and constraints. The systems in between, i.e. at the edge of order and chaos, can exhibit a more flexible and organized behavior. Therefore, it appears likely that self-organisation needs to find a balance between lack of order and too much order."

The edge of chaos phase transition area is then thought to be the locus for evolutionary processes, that involve the perpetual collapse of local structures that then give rise to new patterns of organization, creating a dynamic life-cycle.

Too much order and change will not cross rigid boundaries. Too much chaos and the system loses its organization. Complex adaptive systems - such as ecosystems, societies, and economies - maintain themselves between this randomness and order where they can somehow use both in order to configure and reconfigure themselves, going through both integration and differentiation in evolving to become more complex.

Mitchell Waldrop gives an account of this when he writes: "The edge of chaos is where life has enough stability to sustain itself and enough creativity to deserve the name of life. The edge of chaos is where new idea and innovative genotypes are forever nibbling away at the edges of the status quo, and where even the most entrenched old guard will eventually be overthrown. The edge of chaos is where centuries of slavery and segregation suddenly give way to the civil rights movement of the 1950s and 1960s; where seventy years of Soviet communism suddenly give way to political turmoil and ferment; where eons of evolutionary stability suddenly give way to wholesale species transformation. The edge is the constantly shifting battle zone between stagnation and anarchy, the one place where a complex system can be spontaneous, adaptive and alive."

The idea of the edge of chaos represents a highly abstract but intuitive concept that has come to be applied to many different areas, from business management, to ecology, to psychology political science and various domains of the social sciences.

The idea of the edge of chaos is expressed within the work of the economist Joseph Schumpeter who formulated the idea of creative destruction as the driving force within a market economy. The idea of creative destruction describes how new innovations are constantly being generated by entrepreneurs in order to displace older ones in a continuous cyclical dynamic.

Schumpeter starts in *The Theory of Economic Development* with a treatise of circular flow which, excluding any innovations and innovative activities, leads to a stationary state. This stationary state is, according to Schumpeter, described as the classical economic equilibrium of order and predictability. The entrepreneur is the one that disturbs this equilibrium and is thus the prime cause of economic development, which proceeds in cyclical fashion along several time scales.

Schumpeter contributed to the ideas of evolutionary economics. According to Christopher Freeman, a scholar who devoted much time researching Schumpeter's work: "the central point of his whole life work [is]: that capitalism can only be understood as an evolutionary process of continuous innovation and 'creative destruction.'"

The idea of the edge of chaos has come to be associated with human cognition. When looking at the many possible cognitive states, it is possible to identify the highly predictable and orderly states from those that are more unpredictable and chaotic. In more chaotic regimes, network states are more disconnected from those in the ordered regime. However "at the edge of chaos," the states can be seen to be maximally novel while still connected to states in the ordered regime, and thus are most likely to exhibit the combination of novelty and utility that is the hallmark of innovative thinking. A similar conceptual approach was used to distinguish between chaos, rigidity, and integration to characterize semantic network states in people with Asperger's syndrome, schizophrenia, and healthy semantic processing, respectively.

The edge of chaos hypothesis has been used as a model for studying creativity within individuals. In any system, there are forces pushing towards organization and order and others introducing unpredictability and randomness, a truly creative idea, or creative process, is seen to bridge both of those states. Robert Bilder a psychology professor at UCLA who has studied creativity, says "The truly creative changes and the big shifts occur right at the edge of chaos."

Professor Bilder has tested this by asking children what dimension of a particular learning environment makes them feel most creative. "One of the things they found most valuable in their art classes was the freedom not to have to seek right and wrong answers," Bilder said. "It was that freedom to explore that led them to be increasingly engaged and allowed them to forge connections that allowed them to be more creative." But equally, the creative process also requires some structure in "The ability to inhibit the first thing that comes to mind in order to get to the higher hanging fruit in the cognitive tree is one of the cornerstones of creative achievement."

The edge of chaos hypothesis can be applied to understanding society in terms of the dynamic interaction between micro and macro-levels within social systems. Whereas macro-level social structures, such as laws, religions, governments and other social institutions offer the potential for order and stability within the system, they can also impose too much order on the individuals, limiting their individual development in the name of conformity and group cohesion, ultimately leading to stasis and lack of novelty with which to innovate and evolve the social structure.

Likewise, the micro-level diversity of agent's agents can be seen as a constant source of disorder, pulling in different directions and without macro stable institutions can create the potential for conflict between the many individual agendas of the agents and special interest groups. As Thomas Hobbes famously stated a society in a state of nature without strong political institutions would be "solitary, poor, nasty, brutish and short."

A functioning society can be seen to be one that is able to maintain itself on the edge of chaos with both stable macro institutions that are capable of maintaining sufficient order but can also maintain individual autonomy as much as necessary for the individual to develop. Functioning constitutional democracies that both maintain social order and individual rights through laws, while also providing

mechanisms for the individual members to change those institutions when needed, may be an example of this. As such they enable both an upward and downward interaction in a cyclical fashion that is characteristic of evolutionary process, where new diversity comes from below while constraints and selection come from above to continuously generate new and relevant variants in response to changes in the environment.

Emergent Process

An emergent process is a process of change that involves non-linear, abrupt phase transitions as a system's overall structure and function is transformed into a new regime of behavior, exhibiting new properties that could not have been predicted to arise prior to the transformation.

Emergent processes may be contrasted to linear processes of change. With linear systems, the overall behavior of the system is a direct consequence of the interaction between the parts, at all times. The macro-level features and behavior can be directly computed from the micro-level elementary parts, changes in the micro-level create corresponding changes in the macro-level. For example, micro-level changes in the mechanical parts to a car create the macro-level changes in its position.

Any future or past state to the system can be understood as some combination of changes within the elementary components. We can, for example, predict far out into the future the occurrence of eclipses because the future state of the solar system is governed by the changes of its elementary parts. We can predict into the future where the planets will be and this will be similar in quality to previously experienced states in the past, i.e. nothing qualitatively different will occur.

Many modeling techniques are based upon this assumption of an absence of emergence with the whole process of change being the sum of its micro interactions. For example, within economics dynamic stochastic general equilibrium models(DSGE) attempt to explain aggregate economic phenomena, such as economic growth, business cycles, and the effects of monetary and fiscal policy, as derived from microeconomic phenomena. Linear systems equally show proportionality between input and output meaning that change typically happens in an incremental fashion.

Emergent processes are different from linear processes of change. Systems that involve emergence, i.e. have two different levels of organization, may undergo change on both levels, where changes in the macro-level are not directly correlated to changes on the micro-level. The macro-level undergoes its own processes of change that have their own structure. Whereas with simpler systems the macro-level states were directly correlated to the micro-level with emergence the macro-level states may become disassociated from the micro level to a greater or lesser extent.

For examples ecosystem, societies and economies go through macro-level process of change such as succession and industrialization that have their own dynamics on the macro level, often all the parts have to move together into a new macro level regime and this places a downward cause on the parts to. New macro level regimes within the system emerge when the system converges upon a new set of rules or protocols that drive all of the parts to adopt that new pattern. For example now that we have converged upon digital as the basic format for information encoding there are strong positive externalities of everyone to use digital as analog become increasingly less compatible, thus we get a rapid change from analog to digital as a macro level regime shift, this shift can happen very fast because of positive externalities creating positive feedback. In this way new macro regimes can rapidly emerge within whole economies, societies or technology infrastructure, shifting to a new pattern of organization that would have been difficult to predict beforehand. This period of major, rapid, macro-level change is called a phase transition.

A phase transition is an emergent process of change between different overall states of organization in a system. Phase transitions may be understood as rapid, abrupt transformations in the overall macrostate to the system that is triggered by some small change within an input variable. Thus, whereas with linear change the macro system changes somewhat proportionally in accordance with the change in some input available, with emergent processes this is not so, during

its process of development there are critical stages either side of which its macro structure takes a very different overall makeup given only a small change in the controlling variables.

The simplest example of this is the different phases that water takes as it goes from gas to liquid to solid ice. In the case of the transition from liquid to steam, as temperature increases, this transition happens abruptly when the system approaches the critical value of 100 degrees centigrade at a constant pressure of one atmosphere.

In phase transitions, such as the spontaneous magnetization in ferromagnetism, relations of long-range order emerge in a system under special conditions. Unlike linear changes - which are largely accountable by statistical averages over the micro properties - they are results of the parts working synergistically in a synchronized fashion where unique properties emerge for the whole system that are not such simple averages.

The nature of the phase of the system cannot be related to the microscopic nature of the basic elements composing the system. Thus phase transition processes typically do not require that we understand the micro-mechanisms upon which they rest. It is sufficed to take the system as a whole and from this, we can infer the general phase-transition behavior as a similar macro-level dynamic to that which occurs in all emergent processes. As such system properties do not require derivation from micro dynamics, they are emergent.

When a system undergoes a phase transition, its micro-components get rapidly reconfigured into a qualitatively different macro-structure. However, the properties of the components themselves remain relatively unchanged. Prior and post-macro states correspond to roughly the same configurations of microstates, i.e. phase transitions involve a restructuring of the system on the macro level with only a limited change in the properties of the parts and other local conditions.

This emergent process that engenders phase transitions can be seen in a wide range of systems, including physical, biological and social systems. One example of this might be the outbreak of ethnic violence. The phenomenon of ethnic violence can be seen as a phase transition from a mixed but non-aggressive population of individuals to occasional, abrupt outbursts of widespread conflict. The underlying level of racism and hostility within the members of a society may remain relatively similar before widespread violence breaks out - which represents one macro regime of peace and stability - and as the system flips into a violent outbreak - another regime of widespread violence. Thus it is not that the members have necessarily become more racist and hostile, it is instead that that system was near a phase transition and a small event triggered it to flip from one macro regime to another while the component parts' properties changed only slightly.

Because emergent phase transitions are discontinuous - meaning they go from one overall state to another with limited overlap between them - they involve critical points of change.

These critical points are discrete changes on the macro-level. Critical points occur due to the system having mutually exclusive regimes, thus instead of one regime gradually giving way to another - which would be a continuous change without critical points - what happens when the regimes are mutually exclusive is a discrete and rapid flipping from one into the other, at the critical point. We can see this in the change in dictatorial political regimes. Because they are autocratic - meaning there can only be one macro political regime - a change between regimes has to involve a critical phase transition point; which is what we see empirically, as typically when a regime falls there is positive feedback driving a rapid move towards a new political regime.

Emergent processes of change create qualitatively new systems. Whereas with linear processes of development, the change is within the properties of the parts which can be quantified, however regime shifts change the structure and functioning of the whole which results in qualitatively new behavior and features. For example, if we think about learning a new subject like physics or chemistry, initially one starts out by learning individual bits of knowledge and trying to understand the subject piece by piece within one's existing conceptual framework; thus gaining more knowledge in a quantitative fashion. But at a certain point, one builds up the full repertoire of understanding within a domain to create an integrated conceptual framework. At this point one's way of seeing physical systems will have now changed qualitatively, representing a qualitatively

new conceptual regime. Accumulating more knowledge at this stage would be of limited use, as one now has the overall structure needed to generate one's own knowledge, thus a whole new function has emerged because of the structure of the macro system. Equally, a regime shift has occurred at this stage, from learning to creating knowledge, from student to research.

Equally, this can be seen in learning some new practical skill. When one starts learning or practicing something new one accumulates isolated rules: "do it like this", "if that happens, do this", "don't forget to", etc. These are incremental quantitative changes to behavior because one is essentially just gathering a list of instructions and working through them in an iterative fashion. However, at some point, if successful, these isolated rules will start to become coherent as they coalesce into a new way of acting. At this point one will start seeing them as integrated into some overarching framework, one begins to act from within that overarching framework rather than merely executing on isolated rules. This is a qualitative change as it induces new emergent, macro-level behavior, and functionality.

System Transformation

A transformation is a change of a system from one state to another, this change may be in the component parts of the system, on the micro-level, or a macro-level transformation in the entire system. A micro-level transformation involving a change in a component part to the system can be understood as a linear transformation. In mathematics and logic, a transformation is a process by which one element is converted into another that is equivalent in some important respect but is differently expressed or represented. Linear transformations do not affect the overall system but simply change the internal component parts, according to the rules of the transformation.

A nonlinear transformation is a macro level complete change in the structure and functioning of a system that creates, or it brought about by, a systemic change in its environment. This macro-level nonlinear process of change we would call a systems transformation.

All systems exist within some environment and are defined to a greater or lesser extent by their role or function within that environment. Systems are adapted to a particular environment through their interaction with other systems. A pebble on a beach is the way it is because of its interaction with the sea's motion and other elements around it. A plant is the way it is, because of the ecological conditions under which it has evolved. Likewise, a technology like a USB stick has its structure and function because of its need to interact with other technological elements in its environment.

Systems exchange energy and matter with their environment and through this their overall structure, makeup, and functioning become shaped and defined by that environment. As long as the system stays within the parameters of its environment then it will retain this overall macro structure. Component parts may be changed in response to changes in the environment but this will not have an overall effect in transforming the system. A system is transformed when it goes outside of the parameters of its environment. Changing environments means a qualitatively new set of rules acting on the system and thus the system needing to evolve new characteristics and functions in order to adapt.

A system that exceeds its environmental limits requires a transformation of some kind. The environmental limits may be in space, in energy or other resources required for the system's functioning. All systems that grow faster than their environment will ultimately reach some limit. A classical example of this being exponential growth within a microbial population as they find a new food source; reproduce rapidly leading to exponential growth; ultimately become over populated for the available food source; reach a limit and collapse.

Typically as the limits to an environment are met there is a crisis situation. The system starts to encounter limits providing it with feedback that change needs to occur in order for it to continue. At this stage, the system must either be transformed in order to continue growing, collapse down to a lower level or reduce its operation to that which can be sustained within its normal environment. Though often this last option is not possible as the system has become adapted to operating in an unsustainable fashion, or the environment was only ever transitory. For example, the stages of growth in a human being are transitory. Once it is time to change the context, i.e. be born, become an adolescent, or an adult, then the previous environment ceases to exist. At this stage, there are only two options, one either fails to develop into the new environment or one succeeds in making the transformation, either way, one will not stay within the old context.

Once a system has reached the limits of its environment it starts to receive feedback signals that make it more and more difficult to operate within its current modality. An example of this within a social system would be the stress that an individual experiences when nearing some limit. Such as the stress one might feel from overworking. When we work more than normal stress builds up, the farther away from our normal working regime we go the more the stress builds. At this stage, we

can choose to go back to the previous state by stopping working, or eventually, we will reach some limit and move into a new regime.

If the environmental limit prevents the system from following its long-term development - for example, the person is working very hard because they have to finish a project on time in order to get promoted and continue with their careers - and where the option of returning to a normal state is not possible, then the system must search for a way to transform itself and this is called, exploration of the state of possibilities. This means that the system will "explore" new options, different ways of working and organization because it has found a particular constraint that will not allow it to develop further. For example, the person trying to complete the project may have to re-assess their whole mode of operating i.e. their work processes and way of organizing themselves. If they can find a new system of organization they can achieve a new level of efficiency, reach the next level and in so doing go through a transformation into a new environment.

Emergence plays a central role in a system's transformation, as at a limit the system is forced to create something new, self-organize into a new organizational pattern that is more efficient and thus enables it to operate in a new, broader, more complex environment; or else it must move back to the previous regime or collapses to a lower level. Professor Eve Mitleton-Kelly of LSE talks about this emergent process engendered within transformations as such "what happens is when a system is pushed far-from-equilibrium the following characteristics come into play to create the new order. It will self-organize, it will explore possible solutions, it will co-evolve, new structures will emerge, there will be a sense of coherence, but also the precise behavior can neither be predicted nor controlled."

These macro transformations can be seen in the long-term development of economies and societies. A subsistence agrarian economy can only support a limited number of people within a given area. If a society is to support a higher more dense population than this limit will allow its economic base has to shift into an industrial regime, which represents a whole new macro structure to the technology infrastructure, economic organization, and social institutions. This critical limit between environments represents a phase transition, an unstable region where the system can not maintain itself for long as it is in disequilibrium. At a phase transition limit, a system is typically expending large amounts of energy that can not be sustained for prolonged periods. The current process of industrialization within China is an example, a period of rapid change, with very high growth and energy consumption, wherein most people will be transformed from rural peasants working the land to urban life working in factories and other industrial jobs; a whole new socio-economic modality.

Transformation Theory

Transformation theory is a theory of transformation to structures of change in natural and social systems. The theory illustrates change as a series of successive S-shaped curves, each stage of development within the model engenders three distinct phases of growth and two breakpoints, where the rules at a certain level are discontinued.

Phase One is characterized by experimentation, in which the system attempts to find a connection with its environment. Assuming this connection is found, the first breakpoint is reached. It is at this point that the rules change from experimentation to replication of success. The system must cease searching and begin capitalizing on its connections by simply repeating its formula for success. In Phase Two, the system enjoys major growth, limited only by the environment that provides resources for that growth.

Assuming the system is allowed this ideal growth without unexpected changes, it eventually consumes those resources. At this second breakpoint, the system enters a bifurcation: it begins to open up to innovative changes, to accept information or resources that were explicitly rejected in Phase Two, and it simultaneously reinvents itself. A new S-curve is born at the second breakpoint. For example, when applied to the creative process we can see how it maps onto three distinct phases in the process; invention, improvement, and innovation.

